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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	2	Academic Year :	2026-27

Task - 2

Aim

To implement client-side routing and dynamic state management in React using React Router DOM, configuring declarative routes (/ , /projects, /contact), navigation bars with active state indicators, and reactive state variables (useState, useEffect) for interactive dark/light theme switching.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. BrowserRouter, Routes, and Route configured in App.jsx.
2. NavLink component utilized with isActive styling function.
3. useState hook managing global theme state (Dark/Light mode) persisted in localStorage.

```
// src/App.jsx (Routing & Theme State)
function App() {
  const [isDarkMode, setIsDarkMode] = useState(() => localStorage.getItem('theme') === 'dark');
  const toggleTheme = () => setIsDarkMode(prev => !prev);
  return (
    <BrowserRouter>
      <div className={isDarkMode ? 'dark-theme' : 'light-theme'}>
        <NavBar isDarkMode={isDarkMode} onToggleTheme={toggleTheme} />
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/projects" element={<Projects />} />
          <Route path="/contact" element={<Contact />} />
          <Route path="*" element={<NotFound />} />
        </Routes>
      </div>
    </BrowserRouter>
  );
}
Output
```

Output

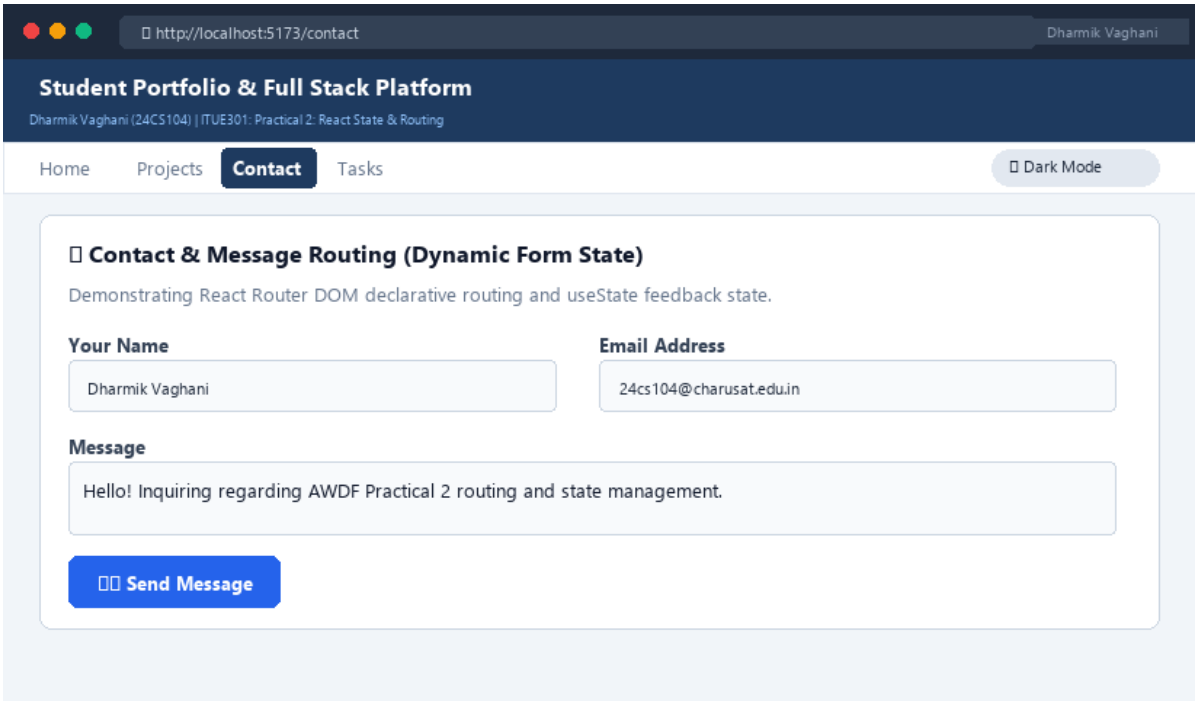


Figure 2: Execution Output & Functional Verification for Practical 2